The Spectrum of a Compact Internal Space

A graphical introduction to the structural and numerical construction

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The framework takes minimal unbiased change and its allowed compositions as the generating principle. The internal construction fixes a compact response space, its mode operations, gauge structure and finite matter carrier. Spectral sums and common source responses then determine quantitative relations among effective couplings, mass scales and macroscopic readouts.

The Spectrum of a Compact Internal Space I: Gauge Structure and a Scalar Mass Gap develops the structural construction. *The Spectrum of a Compact Internal Space II: Effective Couplings and Mass Scales* evaluates its physical consequences with the stated projections, normalizations and matching rules (Guo, 2026d, 2026g). The V5 software records the shared calculation and its comparisons. Newton's constant supplies the observed dimensional anchor.

The derivation chain

The first diagram follows the construction from the generating principle to macroscopic response. Its arrows identify operations in the physical construction. Each stage uses the conditions and physical definitions given in the corresponding manuscript.

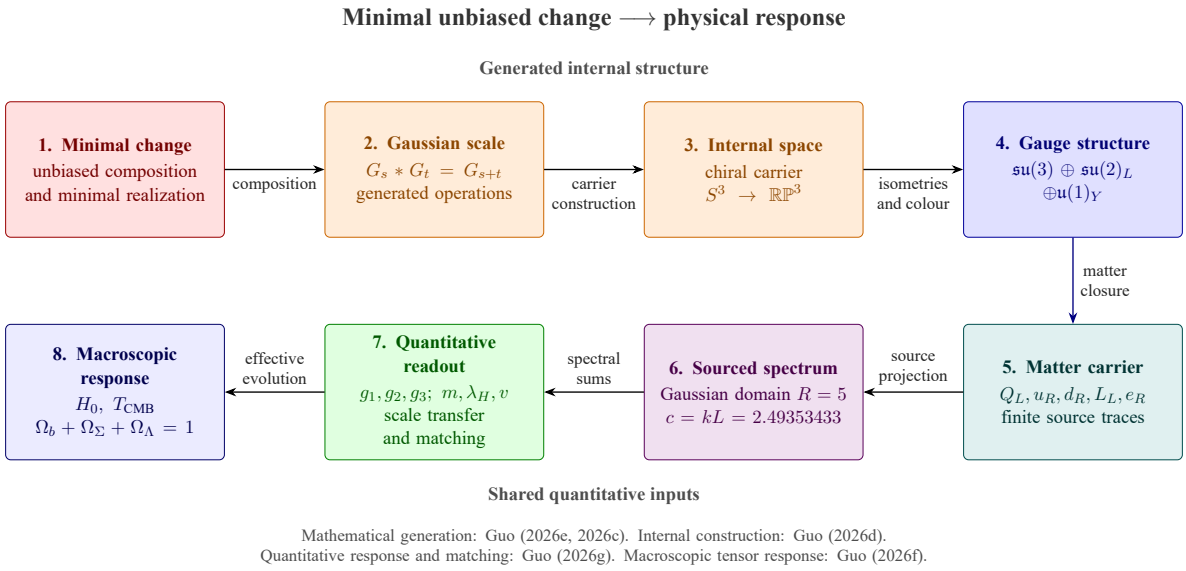


Figure 1. Generated internal structure and shared quantitative readouts.

The common origin is more informative than a list of isolated numbers: the same source normalization and endpoint enter several sectors. Consequently, changes in their common inputs produce correlated changes in couplings, masses and cosmological quantities. These relations provide cross-sector tests of the construction.

Layered structure

The six-layer diagram separates the generating principle, internal objects, source operations and quantitative readouts. The dimensional anchor enters the physical scales through

$$M_P = (8\pi G_N)^{-1/2}, \qquad M_G = \frac{M_P\sqrt{\pi}}{c}, \qquad c = kL.$$

The arrows on the right show the propagation of common inputs into the macroscopic quantities.

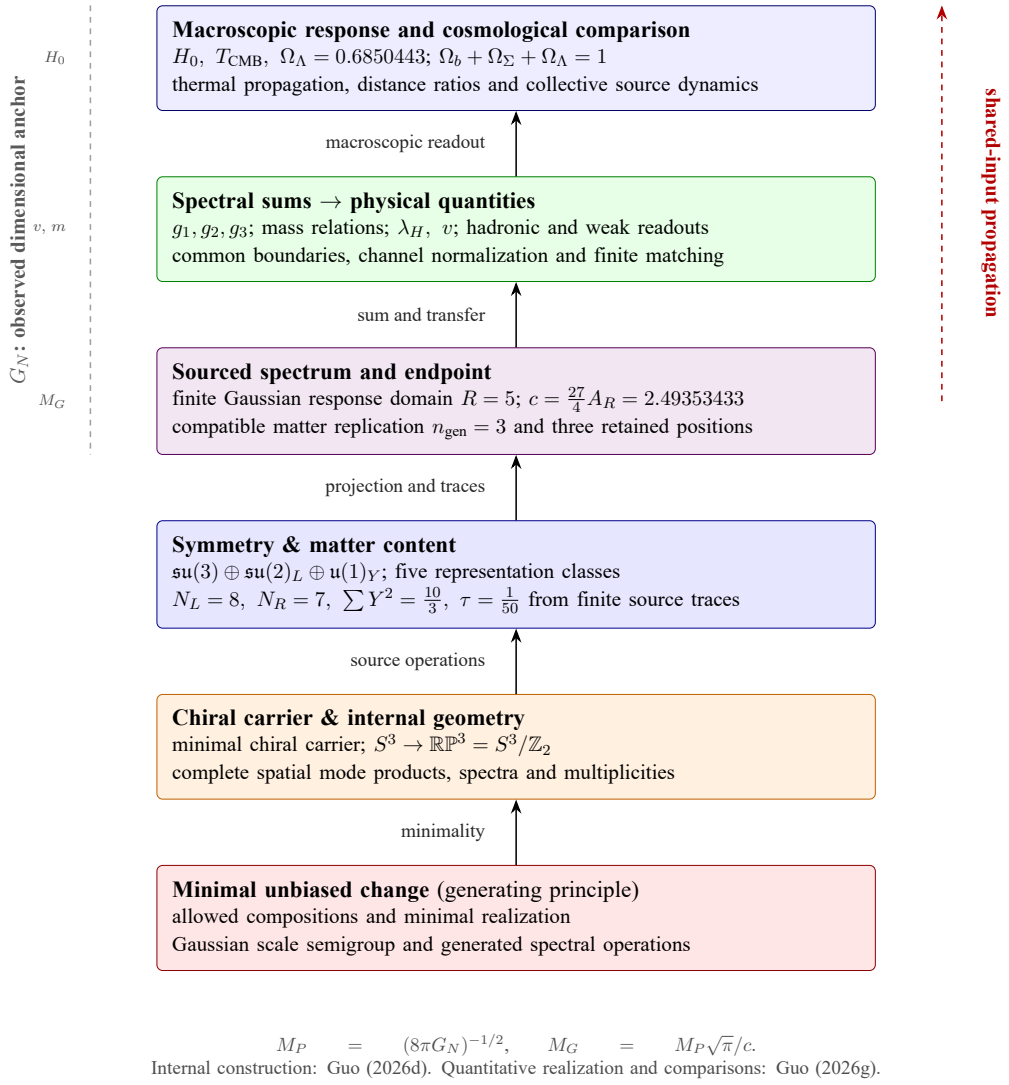


Figure 2. Six layers of the construction and their common scale normalization.

The production calculation uses a Gaussian weight on the finite response domain $R = 5$. At fixed content and response domain, $F_R(c) = A_R/c - 4/27$ has the unique positive root $c_R = 27A_R/4 = 2.4935343325\ldots$. The finite source traces give $\tau = 1/50$, and the joint matter-content and screening conditions select three replications compatible with the retained three-position window.

The complete Gaussian domain defines a different response. At fixed remaining inputs, its electroweak and electron mass readouts are about 10–13% and 3–4% of the respective finite-domain values. This comparison measures sensitivity to the physical response domain; quadrature and spectral-tail errors are controlled separately.

Mathematical structure and common dynamics

The three panels display the maps and identities used in the calculation: spectral projection and matching, Gaussian composition and the endpoint, and the action evaluated on generated variables. The source and its normalization are retained along these maps.

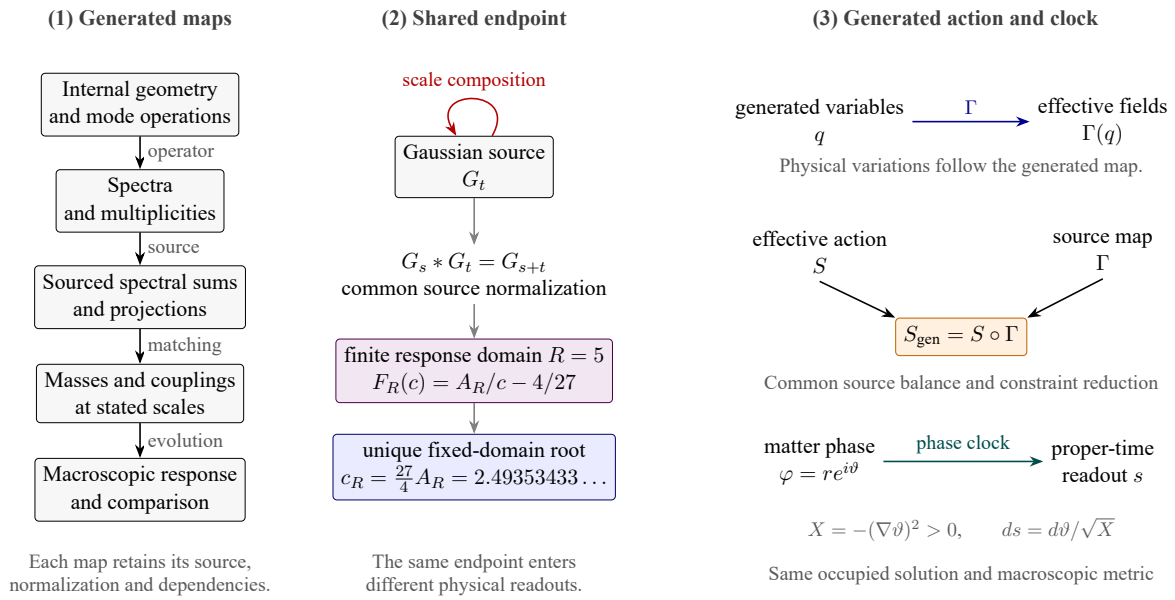


Figure 3. Generated maps, the shared endpoint, and the source–action interface.

Correlated readouts. Changing the bandwidth at fixed remaining inputs links the electron and electroweak scales through

$$\frac{v_R}{v_5} = \frac{c_5}{c_R} \left(\frac{m_{e,R}}{m_{e,5}} \right)^{\pi/5}.$$

The charged-lepton relation $m_\mu^2/(m_e m_\tau) = \exp(\sqrt{2\pi})$ eliminates the common endpoint. Its experimental comparison uses finite matching to the stated running-mass scheme and scale. The independent one-loop combination is $\Delta_L^{(1)} = 0.009337517754$, compared with the correlated requirement $\Delta_L = 0.0112736 \pm 0.0005817$.

A common occupied background. In the auxiliary-radius realization, matter, the collective radius and the macroscopic metric satisfy one action and a common source balance. Constraint elimination gives a positive scalar kinetic form on its certified branch. A continuous certificate covers $0 \le \mu s \le 0.01$; the conserved phase charge of the retained matter pair provides a monotone clock on the same solution. The local nonlinear source–potential system and the expanding finite displacement window have their own stated actions, constraints and error controls (Guo, 2026g, 2026h).

Physical output

The output tree groups current central values by their shared inputs. Mass definitions and approximation orders accompany the nodes. The branching structure is a guide to the calculation; the software records the complete dependencies of each result.

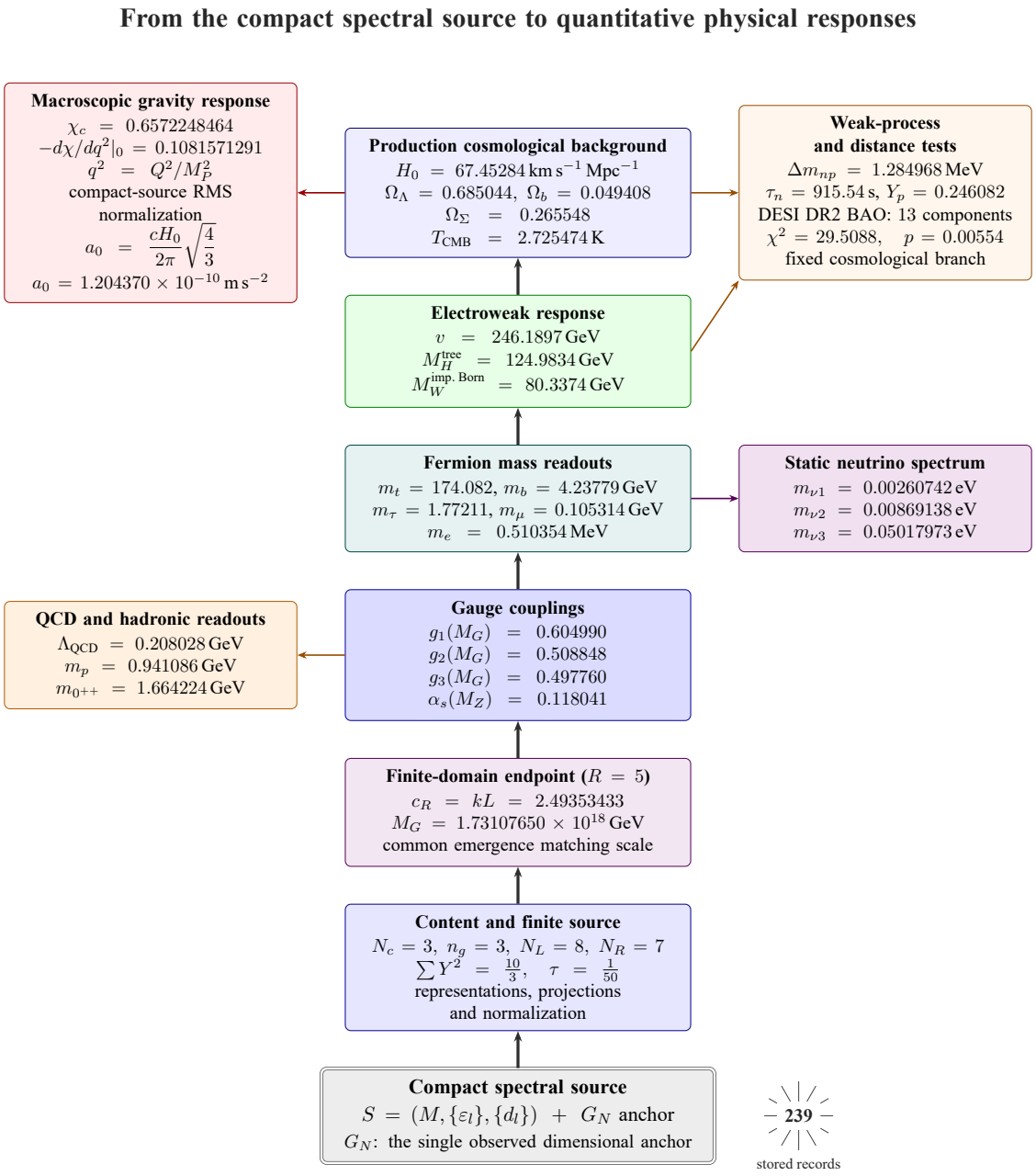


Figure 4. Current V5 readouts and selected joint tests.

Comparison, verification and sources

Central-value agreement is assessed together with the approximation and the shared-input correlations. The neutron lifetime lies 4.23% above its reference; the helium central residual is $+0.44\%$. The fixed thirteen-component DESI distance comparison gives $\chi^2 = 29.5088$ and $p = 0.00553993$, showing tension for the stated thermal background and distance mapping. The direct generated-atomic-input thermal response is reported separately, with $\chi^2 = 27.1191$ and its own finite error budget.

Gravity is represented by a macroscopic collective tensor response. The specified quadratic gauge action supplies two physical massless helicities and a positive source residue. The auxiliary-radius model, local nonlinear response and independent-foliation realization retain their respective constraint and stability results. In particular, the negative longitudinal direction of the independent-foliation static branch remains part of its reported assessment.

The parameter store contains 239 records: 235 derived records, one observed dimensional anchor, one scale choice and two pending entries. They include intermediate quantities and complete responses. Reference observations and covariances occupy a separate store. The software manifest binds source and result fingerprints to the recorded calculation; the verification report lists the numerical, isolation and formal checks actually executed.

References

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